The integral divided norm and the two remaining orbit-twelve compact blocks

D.65. An actual source correction fills two further arithmetic compact blocks [Proved]

Keep the marked sources and all comparison maps of D.64:

$$X = \widehat{B}/(R_{50} \cap L_2), \qquad Y = X/\mathcal{K}, \qquad X_{64} = X/\mathcal{J}_{64}, \qquad \mathcal{L} = \ker(U^- \longrightarrow U_*^-). \quad \text{(D.65.1)}$$

In particular, G is the actual arithmetic group of order $576 \cdot 3^{97}$, and $t = sr$ has order 36 in G. The maps to

$$G^\dagger = (\mathbb{Z}/3)^4 \rtimes Q \longrightarrow G_* \qquad (|G^\dagger| = 46656, |G_*| = 3888) \quad \text{(D.65.2)}$$

are the actual D.60 maps. Use the integral compact lattices M_a, their small maps ϕ_a, and the central modules C_a of D.64. Only $a = 2, 3$, the two orbits of size twelve, enter the new correction.

There is an actual continuous peripheral automorphism of

$$\begin{aligned} X_{65} &= X/\mathcal{J}_{65}, \\ \mathcal{J}_{65} &= \ker\left(\mathcal{J}_{63} \xrightarrow{(q_1,q_2,q_3,\nu_4 q_4)} C_1 \oplus C_2 \oplus C_3 \oplus \nu_4(C_4)\right). \end{aligned} \quad \text{(D.65.3)}$$

It has multiplier one, fixes the original cusp t, is identity on G, and retains the complete D.57 action with its literal small conjugators. Its $\mathcal{W}_h$ action, normalized parameter, and all proved metabelian and CM comparisons through that map are unchanged. Both actual arithmetic branches and all their retained framings are included.

Here are the exact new maps and dimensions. The two D.64 retractions are denoted by $p_a : C_a \to I_a$, with $\dim I_a = 4$. Then

$$\begin{aligned} \mathcal{J}_{65} &\subset \mathcal{J}_{64}, \\ \ker(X_{65} \to X_{64}) = \mathcal{J}_{64}/\mathcal{J}_{65} &\simeq \ker p_2 \oplus \ker p_3, \\ \dim\ker(X_{65} \to X_{64}) = 5752, &\qquad \dim\big(\ker(X_{65} \to X_{64})\big)_G = 12, \\ \dim(\mathcal{J}_{63}/\mathcal{J}_{65}) = 7682, &\qquad \dim(\mathcal{J}_{63}/\mathcal{J}_{65})_G = 26. \end{aligned} \quad \text{(D.65.4)}$$

The action on the lower quotient is the new action described below; its comparison with the chosen D.64 action is retained explicitly.

Let $\mathcal{O} = \mathbb{Z}_2[\omega]$, $\omega^2 + \omega + 1 = 0$. On the actual rank-16 $\mathcal{O}$-lattice M_a, write H_a for the unordered pair stabilizer, of order 48. In integral Hermitian coordinates its exterior action is

$$\rho_a(g)B = \begin{cases} A_g B \bar{A}_g^t, & g \text{ preserves the character}, \\ -A_g \bar{B} \bar{A}_g^t, & g \text{ exchanges the character}. \end{cases} \quad \text{(D.65.5)}$$

Both the whole integral exterior norm and the whole integral endomorphism trace are even. Consequently the maps

$$\begin{aligned} n_a(\bar{B}) &= \frac{1}{2} \sum_{g \in H_a} \rho_a(g)B \pmod 2, \\ k_a(\bar{A}) &= \frac{1}{2} \sum_{g \in H_a} gAg^{-1} \pmod 2, \qquad A : M_a \longrightarrow \ker \phi_a \end{aligned} \quad \text{(D.65.6)}$$

are independent of the indicated integral lifts. The first map descends to the actual coinvariants of the D.64 kernel. Its exact ranks are

a	$\dim(K_a)_{H_a}$	$\operatorname{rank} n_a$	$\dim\ker n_a$	$\operatorname{rank} k_a$
2	10	6	4	16
3	10	6	4	16

(D.65.7)

In each row $\ker n_a$ is exactly the D.64 actual source-involution image. For every permitted finite framing, the actual compact reduction D_a is explicitly determined and is an isometry. With $\Omega_a = \overline{(H_a^{\mathrm{int}})^{-1}}$, the map

$$\bar{A} \longmapsto k_a(\bar{A})\Omega_a \bar{D}_a^{\,t} + D_a \Omega_a \overline{k_a(\bar{A})}^{\,t} \tag{D.65.8}$$

has precisely the rank-six image of n_a. Complete coefficient matrices and right inverses on all ten coinvariant representatives are retained in the certificate. Here H_a^{int} denotes the D.64 intersection matrix after reduction modulo two; Ω^{int} in the integral boundary equation below is the bivector before reduction.

The integral homology changes realizing (D.65.8) factor through actual finite free $\mathbb{Z}_2[G]$-lattices. They have cusp-fixed peripheral derivations $\eta_a : Y \to \mathcal{L}$. Put

$$\beta_Y(g) = (\eta_2(g) + \eta_3(g))\alpha_{64,Y}(g). \tag{D.65.9}$$

The coefficients η_a are selected by the retained right inverse applied to the actual divided-norm cusp error. The proof constructs their two literal source conjugators, at all integral homology levels simultaneously. After the earlier self-block repairs, the resulting Y-action is exactly

$$\mathrm{Int}(b_{t,Y})\beta_Y, \qquad b_t = \prod_j \prod_{i=0}^{35} t^i u_j t^{-i}, \qquad u_j \in S_{57}. \tag{D.65.10}$$

Here only the transported earlier extra-projective self corrections enter b_t. Its Y-image centralizes t, has zero small and $\mathcal{W}_h$ images, and acts trivially on integral homology. The subsequent eight new source-involution corrections and central augmentation corrections leave this Y-action unchanged.

There is also an exact limit on the larger fixed-homology centralizer route. On the actual lattice $\mathcal{L}$,

$$\mathcal{L}^s = (1+s)\mathcal{L}, \qquad \mathcal{L}^t = N_{36}(t)\mathcal{L}, \qquad \ker N_{36}(t) = (1-t)\mathcal{L}. \tag{D.65.11}$$

Thus all homology centralizer changes in the kernel of the literal small comparison are represented by the norm words already tested in D.64. The new lift uses the actual homology change (D.65.9).

The remaining source is explicit:

$$\begin{aligned} \ker(X \to X_{65}) = \mathcal{J}_{65}, \qquad \ker(X_{65} \to Y) &= \mathcal{K}/\mathcal{J}_{65}, \\ \dim \mathcal{J}_{65} &= \dim \mathcal{J}_{64} - 5752, \\ \dim(\mathcal{J}_{65})_{\text{specified small characters}} &= 276478, \\ S_{65} = \{b \in R_{50} : \mathrm{im}_{\mathcal{T}_2}(b) \in \mathcal{J}_{65}\}, \qquad S_{65}/(R_{50} \cap L_2) &= \mathcal{J}_{65}. \end{aligned} \tag{D.65.12}$$

It still maps onto $\ker \nu_4$, of dimension 1918, and is nonzero. The two remaining old-orbit compact norm-kernel coordinates survive the present test. This is a statement about the specified compact test, not a formula for $(\mathcal{J}_{65})_G$. The boundary in the remaining source kernel and the full profinite realization remain unresolved. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The actual compact action and its finite framing. All coefficient modules start from the marked graph in (D.65.2). The original s, r edges, the affine r-translation $(1, 1, 1, 0)$, the recorded free edges, and the original character representatives are retained. On a new character fibre the affine, cusp and compact ranks are $96, 48, 48$ over $\mathbb{F}_4$. The compact decomposition has the original rank-32 projective part and the original rank-16 complement.

The two possible D.57 branches have β-images in G_* with zero translation and linear states 26 and 33. Their literal centralizing lifts in Q are respectively

$$\{476, 498\}, \qquad \{456, 461\}. \tag{D.65.13}$$

The r-conjugator has image one in G, by the retained odd-prime identity component. All later homology corrections lie above that same finite group. Thus its image in $G^\dagger$ is still one. The translation coordinate of β in $G^\dagger$ is zero: the map to G_* is identity on this translation subgroup. These facts give the complete four-element list (D.65.13), including any remaining finite centralizer ambiguity.

The program `d_compact_rigid_source.py` constructs an actual equivariant affine section of the rank-16 compact complement. On the unordered Sylow subgroup of order 16, the cusp lattice is free of rank six over its binary group ring. Six literal regular generators and all their 96 real orbit columns are retained. Extend their coefficient functionals to the affine lattice, coinduce, average over the three Sylow cosets, and average over the odd coefficient-scalar group. The resulting cusp retraction is checked on every column. Subtracting it gives the asserted affine section; all 48 semilinear intertwiners are checked.

For a literal word β with a prescribed image in (D.65.13), the actual edge map is right convolution by

$$e_1 - p(\beta). \tag{D.65.14}$$

Indeed the general source-conjugator edge map is $p(A) - p(\beta) + A_G e_1$, and here $A_G = 1$. Every path term retains its four-dimensional translation phase. The program finds an actual word by breadth-first search on all 46656 marked states, evaluates its full path, applies (D.65.14) to the actual affine section, and then compactifies. All eight results have rank 16, satisfy the marked small square, commute with all semilinear stabilizer maps, and preserve the entire D.64 form.

This determines the compact action of the actual source, including its possible higher-kernel conjugator changes. A difference between two edge maps with the same finite conjugator coordinates factors through the regular edge lattice. Its compact M_a-to-M_a component therefore factors through a projective module. The Higman trace description follows directly by writing the two maps into and out of a regular module in its group basis. On these actual modules every one of the 512 real coefficient columns of that full trace is zero. This is checked in `d_compact_full_trace.py`. Odd-character averaging first reduces a general real map to the recorded coefficient-linear map; the full stabilizer trace is then exactly the checked one. Inflation across the odd kernel of $G \to G^\dagger$ introduces only an invertible odd norm factor. Thus every such difference is zero on the compact reduction. The computed matrices are the actual D_a.

There is a further exact module comparison. Let $H_a^0 \simeq S_4$ be the ordered stabilizer and let $J_a \simeq S_3$ be the subgroup generated by the retained order-three element and its recorded normalizing involution. The fixed subspace of $\overline{M}_a$ has rank eight and maps onto the small rank-four module. A literal section of that map, followed by trace-one semilinear averaging, gives

$$\mathbb{F}_4[H_a^0/J_a] \otimes \overline{P}_{*,a} \xrightarrow{\sim} \overline{M}_a. \tag{D.65.15}$$

The certificate retains the section, all four coset columns, the inverse matrix, every stabilizer action, and both diagonal and off-diagonal blocks of the actual pairing. The small map is the

sum of the four coordinates. This proves the comparison by its maps, and also gives a direct structural check on the zero trace.

The integral graph and the signed exterior lattice. The program `d_compact_integral.py` works in $\mathcal{O}/4$, with the literal encoding $a+4b$ for $a+b\omega$. It starts from the actual D.64 integral cycle lifts. The chosen free-edge rows are identity, so a pure Q-action is obtained by its actual edge permutation and, on an exchanging element, coefficient conjugation. Every resulting cycle identity and every reduction to the old marked matrix is checked. The comparison with the full marked graph of $G^\dagger$ is the literal Fourier map $\sum_v a_{q,v}[q,v] \mapsto \sum_v \omega^{\ell v} a_{q,v}[q]$. On the selected real character pair its inverse has coefficient $3^{-4}\operatorname{Tr}_{\mathcal{O}/\mathbb{Z}_2}(\omega^{-\ell v}u_q)$. Orthogonality of the two nontrivial characters proves both composites, and the actual edge phases make this a chain map commuting with the cusp maps. Thus the twisted graph carries the integral marked module, with its specified coefficient and deck framing.

The cusp columns use the signed word $(sr)^{12}$ in $G^\dagger$, with all twelve translations. A unit-pivot quotient gives the integral compact lattice. The four projective generators are lifted with the order-three eigenprojector and the integral trace-one coefficient $-\omega$, whose sum with its conjugate is one. Coinduction of the lifted eigenfunctionals gives a retraction; correct its composite by its inverse and then apply the same trace-one exchange average. Its reduction is the original D.61 retraction. The complementary columns give the actual integral rank-16 action matrices. All 144 generating group products per orbit are checked modulo four.

These are reductions of actual integral lattices with the stated marking. Projectivity lifts the included projective lattice and its prescribed modulo-four inclusion. Relative injectivity of its dual lifts the prescribed retraction, exactly as in D.61. Correcting the composite by its inverse retains the given reduction modulo four. If two such integral projectors e, e' are chosen, then

$$u = e'e + (1-e')(1-e) \tag{D.65.16}$$

is an equivariant isomorphism, congruent to one modulo two, carrying their two decompositions into each other. Both projective images have zero small image, so the small map satisfies $\phi u = \phi$. Hence these choices are compared by an actual isomorphism preserving the small marking. Conjugating by a matrix congruent to one modulo two does not change either divided map in (D.65.6): its additional terms contain the whole trace or norm, which is already even.

For completeness, the integral Hermitian realization is checked independently on the underlying real exterior lattice. Put $u_i = 1$, $v_i = \omega$ in the i-th coefficient copy. The primitive embedding sends a diagonal coordinate to $u_i \wedge v_i$ and the off-diagonal coordinates to

$$\begin{aligned} 1 &\longmapsto u_i \wedge v_j - v_i \wedge u_j, \\ \omega &\longmapsto u_i \wedge u_j + v_i \wedge u_j + v_i \wedge v_j. \end{aligned} \tag{D.65.17}$$

The coefficients of $u_i \wedge v_i$, $u_i \wedge v_j$, and $u_i \wedge u_j$ give its integral left inverse. Multiplication by $a+b\omega$ has real matrix $\left(\begin{smallmatrix} a & -b \\ b & a-b \end{smallmatrix}\right)$, whereas coefficient conjugation has matrix $\left(\begin{smallmatrix} 1 & -1 \\ 0 & -1 \end{smallmatrix}\right)$. Their exterior actions give exactly (D.65.5), including its minus sign. The independent program `d_compact_integral_exterior.py` checks all 12288 basis images per orbit, with exact encoding and decoding in the ambient rank-496 exterior lattice.

The whole exterior norm is even, as is the whole endomorphism trace. Replacing an integral lift by twice another vector therefore changes the divided reduction by zero. This proves the lift independence in (D.65.6), including for maps to the saturated kernel of ϕ_a. The norm also annihilates every integral augmentation relation, so its divided reduction descends to the claimed coinvariants.

The complete divided correction calculation. The full divided exterior norm has rank twelve on each rank-256 Hermitian module. On the actual rank-240 kernel it has rank six. All 11520 kernel action columns per orbit are checked for invariance. The ten original coinvariant

representatives are retained. Each of the 288 actual source-involution correction columns is killed by the divided norm. Their rank is four, proving exact equality with its kernel.

The full divided endomorphism trace has rank 32. Restricting its input to $\mathrm{Hom}_{\mathbb{F}_4}(\overline{M}_a, \ker \bar{\phi}_a)$ uses all 384 real coefficient columns and gives rank 16. Its outputs still have zero small image. For every actual matrix D_a, the complete linearized bivector columns in (D.65.8) span the same six-dimensional subspace as the divided norm. A right inverse is recorded for each of all ten coinvariant representatives and checked by exact multiplication. Thus the certificate supplies an actual linear section on the entire image, rather than a search for one favorable boundary.

The actual integral source extension. We now realize these coefficient maps in the original source. For an ordinary integral coefficient-linear lift $A : M_a \to \ker \phi_a$, extend it by zero to the other character pairs and compact summands. The surjection in D.64.11 lifts its values into $\mathcal{L}$. Odd-character projection retains the selected orbit. Denote its actual global compact summand by M_a. Coinduction gives a factorization

$$U \xrightarrow{\pi_{M_a}} \mathsf{M}_a \xrightarrow{j} P \xrightarrow{q} \mathcal{L}, \tag{D.65.18}$$

where P is a finite free $\mathbb{Z}_2[G]$-lattice. Explicitly one may use $P = \mathbb{Z}_2[G] \otimes_{\mathbb{Z}_2} \mathsf{M}_a$, with G acting only on its group-ring factor, and $j(v) = \sum_g g \otimes g^{-1}v$. Define q by the chosen ordinary lift on the identity summand and induce it equivariantly. Multiplying q by $3^{-97} \in \mathbb{Z}_2$ removes the normal odd-group multiplicity. On the original character fibre the compact composite is exactly $\mathrm{Tr}_{H_a}(A)$. The first map in (D.65.18) kills every cusp class and all other compact summands; the compact image of the last map is in the selected M_a-kernel. Write $j_U = j\pi_{M_a} : U \to P$ for that first composite.

Write $E = \mathbb{Z}_2[G]$ for the actual Bass–Serre edge lattice. The marked chain sequence is

$$0 \longrightarrow U \longrightarrow E \longrightarrow \mathbb{Z}_2[G/\langle s\rangle] \oplus \mathbb{Z}_2[G/\langle r\rangle] \longrightarrow \mathbb{Z}_2 \longrightarrow 0. \tag{D.65.19}$$

The quotient of E by U is a free $\mathbb{Z}_2$-lattice. A regular group-ring lattice is coinduced; the coefficient-of-one isomorphism identifies maps to it with ordinary lattice functionals. Extend those functionals across this split lattice sequence and coinduce back. This proves that the first composite $j_U : U \to P$ in (D.65.18) extends to a map $\ell : E \to P$. It is the required relative injectivity assertion with its actual extension map, not an assertion that all integral modules are projective.

The actual path cocycle has $p(s) = 0$, $p(r) = e_1 - e_r$. Put $w = \ell(e_1)$. Since (D.65.18) kills the full cusp homology,

$$N_{36}(t)\ell(p(t)) = \ell(p(t^{36})) = j_U([t^{36}]) = 0. \tag{D.65.20}$$

The restriction of P to $\langle t\rangle$ is free. In each regular cyclic orbit, the sum-zero coefficients are exactly the successive differences. This gives $b \in P$ with $\ell(p(t)) = (t-1)b$. Therefore

$$\begin{aligned} &\eta_P(g) = \ell(p(g)) + (1-g)b, \\ &\eta_P(t) = 0, \qquad \eta_P(s) = (1-s)b, \qquad \eta_P(r) = (1-r)(w+b), \\ &\eta_P(n) = j_U([n]) \quad (n \in \ker(\widehat{B} \to G)). \end{aligned} \tag{D.65.21}$$

Composition with q gives the actual η_a. Its restriction to any defining kernel of Y is zero, since that kernel has zero integral homology. Thus it descends to Y. The two coefficients $q(b)$, $q(w+b)$ lift through the actual surjection $S_{57} \to \mathcal{L}$. Conjugating the old images of s, r by those two source elements realizes (D.65.9) literally. Their small and $\mathcal{W}_h$ images are zero, and the same two elements serve at every finite homology level.

The compact change is even, kills all other compact summands, and has zero small image. It also kills all cusp homology. The full homology map consequently has the old invertible diagonal reductions in its cusp/compact filtration. It is an integral automorphism. The full affine change can have an odd cusp component; the proof does not replace it by a doubled affine map.

The integral orientation identity and the actual error. Choose actual full-source lifts of the current two conjugators. Their cusp defect has zero integral abelianization, because the constructed action fixes t on the full homology source. It has an integral class z^{int} in the class-two commutator lattice. Apply the marked compact surface boundary word before reduction. For each cusp, t^{36} contributes the sum of the 36 conjugates of this error. Summing over the literal cusp cosets gives the full group norm. Changes of the cusp conjugator contribute wedges with cusp homology and vanish under compactification. The commutator product on the other side of the boundary word is the integral orientation bivector. Therefore

$$N_G(z^{\mathrm{int}})_{\mathrm{compact}} = T\Omega^{\mathrm{int}}\bar{T}^{\,t} - \Omega^{\mathrm{int}}. \tag{D.65.22}$$

This is the D.55 proof over the integral lattice, with its original oriented cusp convention. It uses no interchange of a finite deck action: the constructed automorphism is identity on G, and therefore preserves the literal character and pair-coset markings. The actual conjugator framings determining its compact reductions are exactly (D.65.13)–(D.65.14).

On the selected induced self block, take the sum of the transported coset coordinates. Under this actual coinvariant map the left side of (D.65.22) becomes $3^{97}N_{H_a}$. This is the full translation norm factor, including the odd kernel in (D.65.2). The reduction of the actual projected orientation is the D.64 $\Omega_a = \overline{(H_a^{\mathrm{int}})^{-1}}$; all odd covering and Fourier factors reduce to one. The modulo-four norm used above is on this actual integral exterior lattice.

Here is the integral comparison with the smaller graph used in the certificate. The kernel of $G \to G^{\dagger}$ has odd order $d = 3^{93}$. On each selected character component the actual pushforward of cover homology is an isomorphism, with inverse $d^{-1}\,\mathrm{tr}$. Indeed pushforward followed by transfer is the sum of these deck transformations, and transfer followed by pushforward is multiplication by d; that kernel acts trivially on the selected component. Compactification commutes with both maps. Adjunction of the compact intersection pairing gives

$$H_G = d^{-1}H_{\dagger}, \qquad \Omega_G = d\,\Omega_{\dagger}. \tag{D.65.23}$$

Consequently the selected version of (D.65.22), in the literal certificate coordinates, is $N_{H_a}(z) = 3^{-4}(T\Omega_{\dagger}\bar{T}^{\,t} - \Omega_{\dagger})$. This retains the full integral odd factor. Here $\Omega_{\dagger}$ is the actual integral bivector transported by the specified Fourier map; only its already verified reduction is needed. Dividing the displayed norm identity by two and reducing gives the comparison used in (D.65.8).

Divide (D.65.22) by two on this block and reduce modulo two. The division on the left is defined on the whole lattice by (D.65.6). For the actual current cusp error write its result as $n_a(q_a z)$. Replacing the compact homology map by $T + \mathrm{Tr}_{H_a}(A)$ changes that result by exactly (D.65.8). The product of the two trace changes is divisible by four and disappears after this particular division and reduction. Only the proved reduction D_a and the actual reduced orientation are used; no unspecified integral compact action is substituted.

Apply the retained rank-six right inverse to this actual value $n_a(q_a z)$, for $a = 2, 3$. Lift its ordinary coefficient matrices to the saturated integral kernel of ϕ_a, and use (D.65.18)–(D.65.21). This defines the actual coefficients in (D.65.9), and makes both divided-norm errors zero. The two character supports are disjoint, so their corrections commute at the displayed linearized level and have no cross term in either selected self block.

Retaining the earlier source and killing the full two blocks. The compact reductions on every summand are unchanged by the new correction. The full compact action still has fixed

left rows on the extra projective summands, is identity there, and is block diagonal on the actual orthogonal decomposition. Thus every earlier injective-norm argument in D.59–D.64 applies with the same maps. The earlier self-block corrections may be reapplied in their actual extra-projective lifts. Their error has an extra-projective factor or a cusp factor, and hence has zero image in either new compact self block. Their cusp words give exactly (D.65.10), including all 36 factors and the actual wrap term. They act trivially on integral homology, so the divided orientation values just computed remain zero. Central augmentation corrections also leave those values unchanged, since the divided norm kills all augmentation relations. This produces a preliminary source lift whose error lies in $\mathcal{J}_{63}$, with $n_a(q_a z) = 0$ for both new blocks.

By the exact kernel calculation (D.65.7), each remaining coinvariant class belongs to the four-dimensional actual source-involution image. Use the four recorded source vectors in each orbit and lift them through $S_{57} \to \mathcal{L}$. The literal word is $u\widetilde{s}(u)$, with the current source involution $\widetilde{s}$. Its error is the actual commutator $[u, \widetilde{s}(u)]$, giving the D.64 map $[e_\Delta(v \wedge sv)]_G$. Its Y-image is fixed by the current involution, so it leaves (D.65.10) unchanged. These eight words kill the entire two ten-dimensional coinvariant errors. Their divided norms are zero by the checked equality of kernels.

After these words, remove any reintroduced moving component in $(1 - e_\Delta)\mathcal{K}$. This is done after the words, using the onto actual two-conjugator augmentation map from D.58 and D.64. Its coefficients lift to $R_{50} \cap L_{*,2} \subset S_{57}$, so they preserve Y, the literal small conjugators, and $\mathcal{W}_h$. Their image in each invariant coinvariant test is zero. Finally apply the same actual augmentation map to the remaining representatives in $C_1 \oplus C_2 \oplus C_3 \oplus \nu_4(C_4)$. Joint surjectivity of the four maps from D.64 supplies the source coefficients. The final error lies in $\mathcal{J}_{65}$.

The defining kernel is invariant: the compact projections kill all cusp terms, the old fixed left rows are retained, the actual M_a-reductions are unchanged and invertible, and their small squares commute. The resulting finite class-two map is an automorphism by its invertible Frattini map. The same two source conjugators give the integral homology automorphism at every level, and retain the prescribed $\mathcal{W}_h$ automorphism. The actual marked fibre-product description of D.58 therefore gives an automorphism of $X/\mathcal{J}_{65}$. This proves (D.65.3), with the literal comparison (D.65.9)–(D.65.10).

The larger fixed-homology centralizer comparison. We give the promised exact maps behind (D.65.11). Restrict (D.65.19) to $\langle s \rangle$. Edge modules and order-three vertex modules are free. The fixed order-two vertices are indexed by $C_G(s)/\langle s \rangle$. The two exact chain sequences identify

$$\widehat{H}^0(\langle s \rangle, U) = 0, \qquad \widehat{H}^{-1}(\langle s \rangle, U) = \ker\left(\mathbb{F}_2[C_G(s)/\langle s \rangle] \longrightarrow \mathbb{F}_2\right). \tag{D.65.24}$$

The map to the small module is the literal map of fixed vertices. Centralizers lift onto across the elementary-abelian odd kernel of $G \to G^\dagger$. Explicitly, a lifted centralizing element has $gsg^{-1} = ks$, with $sk = -k$ in that kernel. Replacing g by $(-k/2)g$ makes it centralize s. At the remaining finite step, $C_Q(s) \to C_{Q_*}(s_*)$ is onto, with orders 16 and 8. The common fixed translation space has nine elements. At the $G^\dagger \to G_*$ stage the actual fixed-vertex map has 72 source vertices, 36 target vertices, and fibre size two. Its reduced augmentation map is onto of rank 35. Every coset and an actual section are retained in `d_source_centralizer_norm.py`. The Tate exact sequence of $0 \to \ker(U \to U_*) \to U \to U_* \to 0$ now gives zero $\widehat{H}^0$ for the kernel. The actual odd-character projector commutes with the kernel map; its direct summand $\mathcal{L}$ has the same vanishing, proving the first equality in (D.65.11).

For the cusp Sylow subgroup of order four, both vertex permutation modules are free. The unique involution has the actual Q-state 314; it is not conjugate to the source state 1, in either Q or Q_*. The full conjugacy classes are checked. Hence it fixes no order-two vertex; order-three vertex freeness follows from the orders. The two chain sequences identify the even Tate group of U with $\mathbb{Z}/4$ and the odd group with zero. The map to U_* is the identity on

the terminal augmentation lattice, so it is an isomorphism on these groups. The kernel is consequently cohomologically trivial for this cyclic subgroup. Averaging over the odd part of $\langle t\rangle$ proves both cusp equalities in (D.65.11). The recorded affine cusp has order twelve in $G^{\dagger}$; its sixth power is state 314 with translation $(0,2,1,0)$. The actual source cusp retains order 36 throughout.

An element of the kernel of the literal small comparison in Y has homology in $\mathcal{L}$. If it centralizes the current involution or cusp, its homology is fixed by the indicated cyclic action. Equation (D.65.11) gives precisely a norm vector. Its actual norm word has that homology; the remaining central difference changes the error only by an augmentation relation. Thus the fixed-homology route has exactly the source maps already tested. The new construction instead has the explicit comparison (D.65.9)–(D.65.10).

The remaining kernel and the execution record. The four maps of D.64 are jointly onto. Replacing the retractions p_2, p_3 by the full second and third coordinates therefore gives (D.65.4), with $2(2880-4)=5752$ and $2(10-4)=12$. The total retained quotient has dimension $1920+2880+2880+2=7682$ and coinvariant dimension $4+10+10+2=26$. The specified residual dimension is $282230-5752=276478$. Joint surjectivity also maps $\mathcal{J}_{65}$ onto $\ker \nu_4$, whose dimension is 1918. This proves the nonzero source-kernel assertion through its actual preimage in (D.65.12).

Eight complete programs retain all finite columns, actual framing words, integral graph maps, inverse matrices, norm relations, linearized equations and correction sections. The separate permutation comparison and real-exterior verification are included. An initial integral run completed the first mathematical block but failed while serializing a variable that had been reused as an index; its source and actual output are preserved. The corrected program uses a distinct section variable. No failed run is counted as a completed certificate. The final copied-source replay and all input hashes are recorded separately. □